

Navigating the future of megaprojects sustainability: a comprehensive framework and research agendas

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Abstract

Purpose – Megaprojects can play a crucial role in achieving Sustainable Development Goals (SDGs) and tackling Global Challenges. However, they are often criticized for their massive resource consumption, poor cost/time performance and significant social-environmental impacts, including irreversible environmental damage. This study aims to chart the evolution of research on megaprojects sustainability and to offer a roadmap for future developments.

Design/methodology/approach – This study adopted a systematic literature review (SLR) to identify emerging themes and gaps. This study follows a three-step process of planning, conducting and reporting, based on the Tranfield approach. The overview of megaprojects sustainability research was captured through bibliometric analysis. In addition, content analysis was carried out to reveal the development of this field and get an insight into the future directions towards sustainability transition.

Findings – This study provides a longitudinal, in-depth analysis of megaprojects sustainability studies. Drawing on sustainability science and project management theories, we introduce a three-dimensional analytical framework consisting of sustainability, scope and stakeholder. This framework explains the evolution of megaprojects sustainability research from sustainability of megaprojects, to sustainability for megaprojects and then to sustainability by megaprojects. Three future avenues are proposed: (1) SDGs orientation at multi-level; (2) scope enlargement at temporal and spatial scales and (3) inclusive development for stakeholders.

Originality/value – This research contributes to the literature by providing a comprehensive and forward-looking analysis of megaprojects sustainability research. The framework and three research agendas provide a comprehensive picture of megaproject sustainability research; the agenda for future research is intended to inspire more studies and disruptive actions towards sustainability transition.

Keywords Megaprojects, Sustainable development, Sustainability transition, Systematic literature review

Paper type Research paper

1. Introduction

Humanity faces unprecedented challenges, such as climate change, poverty eradication and social injustice (Wu *et al.*, 2022a, b). According to the United Nations (2022, 2023), only 12% of the SDG targets are on track, with nearly half showing weak and insufficient progress. One-third of the SDGs, including poverty (SDG 1), hunger (2), education (4) and health (6), have seen no improvements or have regressed below the 2015 baseline. Achieving the SDGs by 2030 seems unrealistic. Business-as-usual approaches are no longer sufficient, as stated by the United Nations (2023): “We cannot simply continue with more of the same and expect a



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different result". Urgent transformations and disruptive actions are required. Megaprojects, regarded as socio-technical efforts in complex institutional environments, provide substantial big-push solutions that can advance (or hinder) sustainability transition and SDGs (Flyvbjerg, 2017; Papadonikolaki *et al.*, 2023; Whyte and Mottee, 2022; Winch, 2022).

Megaprojects are vast, highly complex and uncontrollable ventures requiring large investments, involving a plethora of stakeholders and having transformative global impacts (Flyvbjerg, 2014; Pitsis *et al.*, 2018). Megaprojects are crucial pillars of modern societies, driving national economic growth, social production and improved quality of life (Clegg *et al.*, 2017; Flyvbjerg, 2014; Guikema, 2009). Infrastructure is particularly prominent in megaprojects and has potential impacts across all 17 SDGs (Thacker *et al.*, 2019). Remarkably, megaprojects also have negative impacts that often extend far beyond their immediate scope and lead to enormous and irreversible consequences (Ma and Fu, 2022). For instance, large hydropower and mega urban development projects often cause long-term social controversies, displacement and environmental degradation, exacerbating social inequalities (Khanal *et al.*, 2021; Wang and Wu, 2019). Major coal mining megaprojects can have ripple effects on ecosystems, posing direct and deadly threats to marine life and cascading impacts on geological stability and fisheries (Saint-Amand *et al.*, 2022). There will be increasingly complex megaprojects worldwide in the foreseeable future (Ma and Fu, 2022; Thounaojam and Laishram, 2022). To achieve ambitious sustainability visions, future megaprojects will face enormous and transformational challenges at multiple levels, including institutional, policy and organisational levels (Sovacool *et al.*, 2022).

Against this background, it is essential to rethink megaprojects sustainability. Existing studies on megaprojects sustainability are scattered across epistemological communities and often focus only on certain types of megaprojects or specific aspects of sustainability. Given the relevance of this topic, comprehensive investigations have been published recently. Thounaojam and Laishram (2022) examined key sustainability issues in mega infrastructure projects. Wang *et al.* (2020) conducted a scientometric analysis investigating keyword co-occurrence and cluster analysis. Yet, even these recent work did not intend to holistically capture the dynamic evolution and future trajectory of megaproject sustainability studies. Therefore, this paper fills the gap by systematically aggregating and analysing the existing body of knowledge through a systematic literature review (SLR) approach. The ultimate aim of this paper is to provide a holistic perspective on the development of megaprojects sustainability literature and a roadmap for future research.

The remainder of this paper is organised as follows. In the next section, the methodology is detailed. This includes a SLR and a bibliometric analysis of the relevant scientific literature. In the third section, a three-dimensional research framework is introduced, from which three research agendas are derived to summarize the development process of megaprojects sustainability research. Further, this paper reports the key directions from sustainability, scope and stakeholder dimensions and provides a research agenda for the future. Finally, the last section concludes this study.

2. Methodology

SLR is an efficient and high-quality method to advance scientific knowledge by exploring literature, identifying emerging themes and gaps in a clear and replicable manner (Cook *et al.*, 1997; Mulrow, 1994; Tranfield *et al.*, 2003). We followed a three-step process of planning, conducting and reporting on articles, as recommended by Tranfield *et al.* (2003) and widely used in management research (e.g. Crossan and Apaydin, 2010; Soundararajan *et al.*, 2024; Stefano *et al.*, 2023; Thounaojam and Laishram, 2022). First, a review protocol was created to provide a clear scientific research strategy (Tranfield *et al.*, 2003), identifying the research question, the database, keywords, screening conditions and inclusion and exclusion criteria. Second, relevant articles were identified and selected, and data were extracted, synthesized and analysed. Lastly, the findings were reported and discussed, including the state of the art

and insights into gaps and future directions. Table 1 summarizes the review protocol and the whole process.

2.1 Search strategy

The first important decision is establishing which scientific repositories to use (Neto *et al.*, 2016). We selected the Web of Science (WoS) core collection because it is widely regarded as a comprehensive and authoritative data source (Neto *et al.*, 2016; Olawumi and Chan, 2018). The search terms were carefully defined and pretested to ensure rigor and completeness. The key topics of this research are megaprojects and sustainable development. We first identified the initial search terms based on building blocks (Booth, 2008). We started with an extensive review of previous literature reviews to learn which terms were used in papers already published in leading journals. To date, there is no strictly consistent definition of megaprojects, especially in terms of cost thresholds (Hu *et al.*, 2015). However, there is a strong consensus on the characteristics of megaprojects that they are large-scale, complex with long-lasting impacts and involve many stakeholders (Babaei *et al.*, 2023; Brookes and Locatelli, 2015; Denicol *et al.*, 2020; Flyvbjerg, 2014; Li *et al.*, 2018; Wang *et al.*, 2017, 2020; Wiewiora and Desouza, 2022). In addition, despite being many categories of megaprojects, the literature is predominantly on infrastructure megaprojects (Flyvbjerg, 2014; Thounaojam and Laishram, 2022; Wang *et al.*, 2018). Therefore, consistently with the aforementioned studies, multiple variants and synonyms of megaprojects were used for searching. As reported in Table 1 the exact portion of the string used in our search is: “(*megaproject\$ or mega-event\$ or (mega or major or large or large-scale or vast or public or complex) NEAR/2 (project\$ or infrastructure) or ((infrastructure or construction or engineering) NEAR/2 project\$)*)”.

Regarding sustainability, general terms such as “sustainability” and “sustainable development” are the most used in the existing literature, and social, economic and environmental aspects are widely recognised dimensions. We identified a wide range of related terms such as “econ*”, “financial”, “cost”, “performance”, “social”, “stakeholder”, “responsibility”, “corrupt*”, “environment”, “green” and “pollution” after snowballing an extensive review of relevant literature reviews (e.g. Aarseth *et al.*, 2017; Caiado *et al.*, 2017; Sabini *et al.*, 2019; Wang *et al.*, 2020) and brainstorming among the authors. Also, as suggested by Kuhrmann *et al.* (2017), multiple trial-and-error searches were also conducted to finalize the search strings for the formal retrieval. The search strings were generated by merging all terms with a combination of relevant Boolean rules. As reported in Table 1 the exact portion of the string used in our search is: “(*sustainability or sustainable or financial or fiscal or econ* or cost or performance or social or societ* or responsibility or stakeholder or accountability or ethic* or citizen* or corrupt* or immigrant or environment* or green or eco* or pollution)*”. We used the “AND” operator to link this string with the string related to megaprojects.

We selected “article” and “review article” in English from peer-reviewed journals in the WoS core collection. The publication year was not limited so that we could collect evidence of the origins and evolutionary trends of this field. The initial search yielded 23,659 documents. The authors hand-screened the articles according to inclusion and exclusion criteria (see details in Table 1), following the principles of full-text availability and relevance to the study (Caiado *et al.*, 2017). The analysis of article titles, abstracts and keywords yielded 843 papers for full-text analysis. Ten papers were excluded due to unavailability, withdrawal or retraction, leaving 833 papers for the full-text analysis. The authors worked on full-text analysis independently and discussed it regularly. Through this process, 316 papers were excluded for being off topic. Thus, a total of 517 articles informed the review and served as the basis for bibliometric analysis.

Stage	Description
<i>Stage I - PLANNING</i>	
Identification for the need for a review	<i>Establishing the Research Objectives</i> • Reveal the development process of megaprojects sustainability research and the state of the art (what we know) • Provide insights into future directions of this field to respond to grand challenges of sustainable development (what we should know)
Preparation and development of a review protocol	<i>Selecting electronic search in databases</i> • Web of Science (WOS) Core Collection <i>Selecting Search Terms based on building block technique</i> • (megaproject\$ or mega-event\$ or ((mega or major or large or large-scale or vast or public or complex) NEAR/2 (project\$ or infrastructure)) or ((infrastructure or construction or engineering) NEAR/2 project\$)) AND (sustainability or sustainable or financial or fiscal or econ* or cost or performance or social or societ* or responsibility or stakeholder or accountability or ethic* or citizen* or corrupt* or immigrant or environment* or green or eco* or pollution) in Topic <i>Identifying filters</i> • Journals: Peer-reviewed academic journals • Type: Article or Review Article or Early Access • Language: In English • Time frame: no bottom limited • Research Domains/Areas: related to project management and sustainability (e.g. Construction Building Technology, Development Studies, Economics, Energy Fuels, Engineering Civil, Engineering Environmental, Engineering Industrial, Engineering Multidisciplinary, Environmental Sciences, Environmental Studies, Geography, Green Sustainable Science Technology, Management, Multidisciplinary Sciences, Regional Urban Planning, Transportation, Transportation Science Technology, Urban Studies) <i>Identifying the Inclusion and Exclusion Criteria</i> • Including the article has at least one pair of search keywords appear in Title/Abstract/Keywords • Including the article focuses on or aims at contributing to megaprojects sustainability research • Excluding articles with no full text available • Excluding the article that the research objective is related to sustainability but sustainability is not the core contribution of the article • Excluding the article that focuses on sustainability but not in the context of megaprojects
<i>Stage II - CONDUCTING</i>	
Identification and selection of studies	<i>Searching in WOS Database</i> • N = 23,659 <i>Screening according to the Inclusion and Exclusion Criteria</i> • N = 517
Study quality assessment	<i>Assessing the quality and relevance of articles for eligibility</i> • Filter all articles based on metrics: (i) articles published in journals with IF over 4.5, (ii) articles cited 20 times or more, (iii) recent articles (2020–2022) • Targeted review on major project management journals • Full-text review of the selected articles to evaluate the alignment between research methodology and research questions • N = 215

Table 1.
The review protocol
and research process of
this study

(continued)

Stage	Description
Data extraction and synthesis of studies	<i>Extraction and coding</i> • Construct the analytical framework and develop a codebook on Microsoft Excel • Extract descriptive information (e.g. title, authors, journal, year) • Full-text analysis and coding of the selected articles: (i) coding all articles for the research context, aims, findings and contributions; (ii) coding all articles based on the analytical framework of three dimensions, including sustainability goals, project life-cycle phase and stakeholders
<i>Stage III - REPORTING</i> Report and discuss implications	<i>Bibliometric analysis</i> • Year of publication • Journal distribution • Keyword cluster <i>Content analysis</i> • Three research agendas: <i>Sustainability of megaprojects, sustainability for megaprojects, sustainability by megaprojects</i> • Gaps and future research agenda

Source(s): Created by author

Table 1.

To ensure the quality of the content analysis and capture the core research trends, the 517 selected articles were screened further, deciding to bring forward the article that satisfies at least one of the following criteria (Crossan and Apaydin, 2010; Hanelt *et al.*, 2021): (1) articles published in journals with impact factor (IF) over 4.5 (Crossan and Apaydin, 2010; Zhou *et al.*, 2021); (2) considering citations as a proxy of quality (Saha *et al.*, 2003), articles cited 20 times or more as recommended by Hanelt *et al.* (2021), (3) considering citation biases and time lags, recent articles (2020–2022) published in journals with impact factor (IF) over 4.5. Additionally, to prevent the omission of relevant field-specific insights, targeted reviews were conducted on major project management journals, specifically the *International Journal of Project Management*, *International Journal of Managing Projects in Business* and *Project Management Journal*. The authors eliminated inter-group duplications and further reviewed the full text of the selected articles to evaluate the alignment between research methodology and research questions (Tranfield *et al.*, 2003). Through iterative discussions, 215 key articles were finally identified as the focus of the content analysis.

2.2 Data extraction and synthesis

Microsoft Excel was used to create a structured database as recommended (Tranfield *et al.*, 2003). A codebook adapted from Laplume *et al.* (2008) was applied to collect data for bibliometric and content analysis. First, descriptive information (e.g. title, authors, journal, year) was extracted from the WOS database. We analysed the literature bibliometrically considering year of publication, journals and keyword clusters using Python and VOSviewer R1.6.19. Section 3 reports the results.

Second, we conducted a content analysis of the 215 papers identified. After an in-depth review of each paper in the sample, we began by thoroughly coding all articles for the research context, aims and objectives, findings and contributions. To effectively capture the themes and trends evolving within the field, we constructed an analytical framework combining sustainability science and project management theories (Clark and Harley, 2020; Metson *et al.*, 2012; Silvius and Schipper, 2014). This framework, detailed in Section 4, consists

of sustainability goals, scope and stakeholders. It guided the coding process of each article, providing a structured and repeatable approach. To minimize subjective bias, three independent researchers conducted iterative analysis and coding of all papers, convening periodically to discuss, consult and achieve consensus. The data extracted laid the foundation for a comprehensive analysis of the literature, facilitating the identification of prevailing trends and patterns. Table 2 provides a comprehensive description of our codebook.

3. Bibliometric results

3.1 Year of publication

Figure 1 illustrates annual publication trend in megaprojects sustainability research. The field emerged in 1977 with Chalmers’ (1977) study on the impact of large energy-related construction projects on surrounding communities. This almost coincides with the rise of sustainable development, exemplified by the United Nations Conference on the Human Environment in 1972. However, research activity remained scarce until the early 2000s, with fewer than 5 publications on average between 2001 and 2007. Starting from 2008, publishing in this area accelerated. In particular, 2015 saw a surge in articles, twice as many as the previous year. The number of articles peaked in 2020 at 65, and by mid-2022, the count at 49 articles has already surpassed that for 2019 (42). In summary, although megaprojects sustainability gained attention alongside sustainable development, research in this field lagged behind at least for the early three decades. Over the last two decades, however, there has been a growing interest, which is expected to continue in the future.

3.2 Journal distribution

Figure 2 illustrates the top 20 journals publishing studies on megaprojects sustainability. The journals publishing the most are *Environmental Impact Assessment Review* (21) and *Journal*

Code	Definition of code
<i>Quantitative variables coded</i>	
Article title	Title of the article
Authors	List of authors
Year	Year of publication
Journal	Publication in which the article was published
Keywords	Author keywords of the article
Methodology	Qualitative, Quantitative, mixed methods
Project	Type of megaproject
<i>Qualitative variable coded</i>	
Research questions	Research questions explicitly stated in the article
Aim of the article	Main aims and objectives of the article
Findings	Key findings explicitly stated in the article
Contributions	Contributions explicitly stated in the article
Megaprojects context	The type of megaprojects and spatial context that the article concerned
Project lifecycle	Some megaproject phase (initiation, planning and design, construction, operation), the whole lifecycle, extended lifecycle (e.g. lifecycle of delivery and its artefacts)
Sustainability concern	Economy, society, environment, multi-dimensions, others (TBL, SDGs, ESG, etc.)
Stakeholder involved	Internal stakeholder, external stakeholders, both, broader society

Table 2.
Codebook for the
content analysis of
this study

Source(s): Adapted from Laplume et al. (2008)

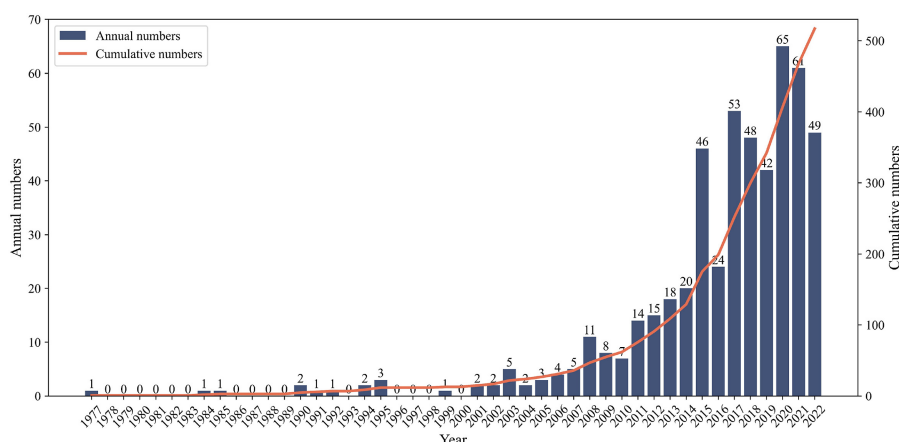


Figure 1.
Number of articles
published annually

Source(s): Created by author



Figure 2.
Ranking of top 20
journals publishing
megaprojects
sustainability articles

Source(s): Created by author

of *Cleaner Production* (21), both covering environmental and social science topics. Project management and construction journals make a significant contribution: *Journal of Management in Engineering* (16), *International Journal of Project Management* (15), *Journal of Construction Engineering and Management* (13), *Engineering Construction and Architectural Management* (11), *Automation in Construction* (6) and *International Journal of Managing Projects in Business* (5). Overall, environmental science, engineering and management dominate the research landscape in this field.

3.3 Keyword clusters

Figure 3 shows the keyword co-occurrence network and clusters. The labels for each cluster were extracted from keywords, and top terms are the most representative keywords that have a high frequency of occurrence in each cluster. The keyword clusters are numbered in

descending order considering the number of items, beginning with the largest #1. As shown in Figure 3, there are 13 clusters, including social sustainability, mega-event, climate change, sustainability, governance, sustainability assessment, stakeholders, sustainable development, megaprojects, project management, Belt and Road Initiative (BRI), infrastructure and indicators.

Sustainability (cluster #4), sustainable development (cluster #8) and megaprojects (cluster #9) highlight the centrality of these themes in the current body of literature. Mega-events (cluster #2) and infrastructure (cluster #12), especially the BRI, are the scenarios of greatest concern in this field. Second, social sustainability (cluster #1), climate change (cluster #3), governance (cluster #5) and stakeholders (cluster #7) constitute mainstream research. Notably, governance-related keywords are often associated with the ecology (e.g. green infrastructure, ecosystem services) and stakeholders (e.g. stakeholder management, public participation). Therefore, environmental and social aspects gain the most attention. Prominent “social themes” include stakeholder management, public participation and social responsibility, while the “environmental themes” include climate change, and environmental impact assessment. Furthermore, cluster #6 (sustainability assessment) and cluster #13 (indicators) indicate that sustainability assessment, indicators and decision-making issues are a research focus in this field.

4. A systematic framework and agendas of megaprojects sustainability

In conducting the content analysis, we used a three-dimensional analytical framework (see Figure 4), leveraging the sustainability and project management literature. This section presents the analytical framework, followed by the main themes of megaprojects sustainability refined through an inductive content analysis of the literature, and categorized into three research agendas: *sustainability of megaprojects*, *sustainability for megaprojects* and *sustainability by megaprojects*.

4.1 A three-dimensional analytical framework

In the sustainability literature, the goals of sustainable development occupy a central place, including *what is*, *what was* and *what should be*. Specific goals depend on stakeholders as well

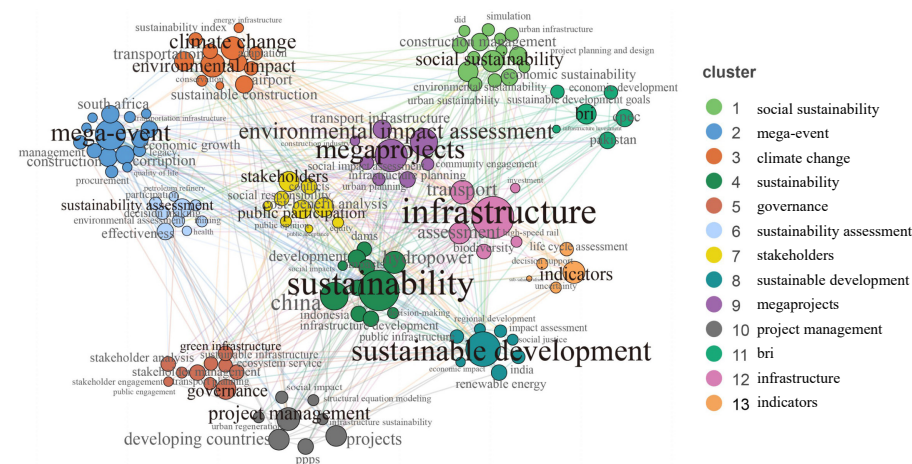
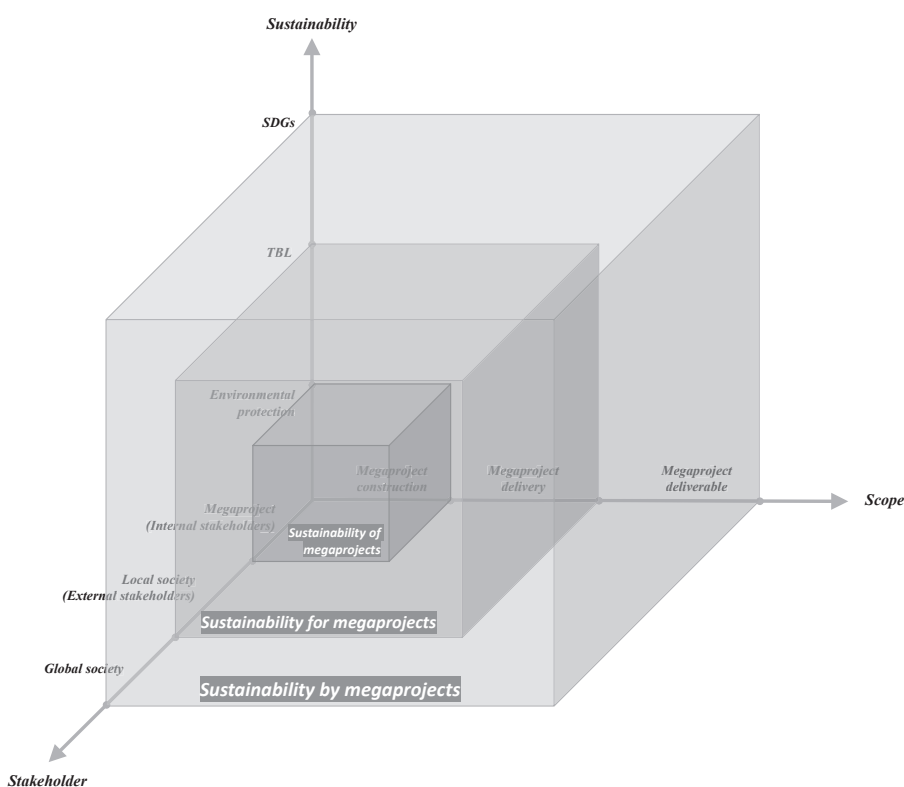


Figure 3.
Keyword clusters of
selected articles in
megaprojects
sustainability
literature

Source(s): Created by author



Source(s): Created by author

Figure 4.
The framework and
research agendas of
megaprojects
sustainability

as temporal and spatial conditions (Clark and Harley, 2020). Stakeholder is one of the fundamental element of sustainable development (ISO, 2010) and together with sustainability goals, they are two core elements of the nature-society interactions at meso level (Clark and Harley, 2020). In addition, a holistic understanding that includes spatial and temporal conditions is critical to exploring the elements necessary to achieving sustainable development and avoiding unintended consequences (Metson *et al.*, 2012).

In project management, sustainability means moving beyond traditional practices to encompass economic, social and environmental impacts (Silvius and Schipper, 2014). Inclusive planning processes across stakeholders, temporal and spatial scope are essential elements (Arts, 2007). Therefore there are four dimensions in sustainable project management: the triple bottom line (TBL) of environmental, economic and social aspects, the temporal and spatial life-cycle dimension, the stakeholder dimension and the values dimension (Silvius and Schipper, 2014). The concept of value is fundamentally embedded in sustainable development (Silvius *et al.*, 2017; United Nations, Department of Economic and Social Affairs, 2015). These dimensions are also widely recognised in the megaprojects sustainability literature (e.g. Wang *et al.*, 2020; Zeng *et al.*, 2015).

Therefore, we developed an analytical framework consisting of sustainability, scope and stakeholder dimensions to make sense of the megaprojects sustainability literature and research. The term *scope* here is used to refer to the temporal and spatial dimensions of the megaproject life-cycle.

In this framework, the core dimension is sustainability goals. Environmental protection has historically been a key aspect of sustainable development (Folke *et al.*, 2016) and is the earliest one incorporated into megaprojects performance. Since the release of the Brundtland report (WCED, 1987) and the introduction of TBL by Elkington (1997), the three pillars of economy, society and environment have become the most recognised models of sustainable development. SDGs now form the most pervasive global framework for sustainable development (Biermann *et al.*, 2022), with the relationship between the three pillars evolving from intersecting to encompassing (Folke *et al.*, 2016; Zhu, 2017).

Regarding scope, megaprojects life-cycle usually includes initiation, planning and design, construction, commissioning and operation (Zeng *et al.*, 2015). The construction phase was traditionally the focus of project scholars, but since the actual impacts are often permanent and inevitably extend beyond the scope and location of implementation (Bojorquez-Tapia *et al.*, 2022), involving global supply and value chains (Cleland and Gareis, 2006; Silvius *et al.*, 2017), the focus is more shifting toward long terms consequences. Therefore, sustainable megaproject management considers not only the life cycle of the megaproject (delivery), but that of the artefacts it creates (deliverable) (Labuschagne and Brent, 2005; Silvius *et al.*, 2017; Silvius and Schipper, 2014).

Stakeholders collectively shape megaprojects' outcomes (Daniel and Daniel, 2019). Megaprojects involve complex self-organising networks of numerous interacting stakeholders (Chapman, 2016; Daniel and Daniel, 2019; Ruuska *et al.*, 2011). The range of stakeholders changes dynamically with different project phases and contexts. They usually include internal stakeholders (e.g. owners, contractors, designers) and external stakeholders (e.g. local communities, pressure groups, the general public) (Zeng *et al.*, 2015). In addition, as megaprojects can have global impacts, the broader society beyond the project context is also considered a stakeholder.

The three dimensions are intrinsically interrelated, especially when megaprojects are considered as complex adaptive systems. Throughout the life-cycle of megaprojects spanning years or even decades (Brookes *et al.*, 2017), sustainability goals and stakeholders evolve dynamically across phases and time points (Zeng *et al.*, 2015).

4.2 Three research agendas

From Agenda 1 to Agenda 3, these three agendas describe the development process of megaprojects sustainability research, accompanied by the connotation of the three dimensions and the continuous enrichment of research themes in the field, as shown in Figure 4 and Table 3. In general, the development of megaprojects sustainability research has followed the evolution of sustainable development concepts.

4.2.1 Sustainability of megaprojects. Since the 1970s, environmental protection has gained prominence in scientific literature and discussions. Early literature on sustainable development primarily addressed environmental aspects (Haughton and Counsell, 2004). Agenda 1 *sustainability of megaprojects* focuses on delivering megaprojects that minimize environmental impacts, primarily concerning construction. Popular themes include incorporating sustainability principles into project performance, assessing environmental sustainability and implementing green practices such as green procurement and green construction.

Incorporation of sustainability principles. Incorporating sustainability principles has expanded success criteria beyond the traditional "iron triangle". In the 1990s, the World Bank advocated for including environmental issues in the economic dialogue (Goodland, 1990; O'Riordan, 1990). Factors beyond economic performance, such as resource efficiency, safety, stakeholder satisfaction and conflict reduction, have become increasingly critical in megaprojects (Toor and Ogunlana, 2010). In particular, environmental considerations have

Agenda	Theme	Exemplary studies
Sustainability by megaprojects	Sustainability of megaprojects	Incorporation of sustainability principles
		Assessment of environmental sustainability
		Green procurement
	Green construction	Goodland (1990), Ozcan- Deziz and Zhu (2017), Toor and Ogunlana (2010)
		Harris <i>et al.</i> (2009), Saxe <i>et al.</i> (2020), Zhou and Sheate (2011)
	Sustainability life- cycle assessment	Ahn <i>et al.</i> (2013), Liang and Chong (2019), Metham <i>et al.</i> (2022)
		Iligan and Irga (2021), Liu <i>et al.</i> (2017), Shang <i>et al.</i> (2021)
	Sustainable design	Keller <i>et al.</i> (2015), Müller <i>et al.</i> (2021), Shen <i>et al.</i> (2012)
		Lauermann (2020), Liu <i>et al.</i> (2018), Xue <i>et al.</i> (2022)
	Sustainable procurement	Brooks and Rich (2016), Ogunsanya <i>et al.</i> (2021), Uttam and Roos (2015)
Sustainability for megaprojects	Social sustainability assessment	Strauch <i>et al.</i> (2015), Uttinger <i>et al.</i> (2005), Xiaer <i>et al.</i> (2022)
		Di Maddaloni and Davis (2017), Li <i>et al.</i> (2016), Prayag <i>et al.</i> (2013)
	Stakeholder management	Ma <i>et al.</i> (2021), Wang <i>et al.</i> (2017), Zeng <i>et al.</i> (2015)
		Social responsibility
	Anti-corruption	Nunkoo <i>et al.</i> (2018), Owusu and Chan (2019), Wang <i>et al.</i> (2017)
		Ansar <i>et al.</i> (2016), Chakrabarti (2018), Hao and Cho (2022)
	Sustainable economic growth	Annear <i>et al.</i> (2021), Browne and Lowe (2021), Freelove and Gramatki (2022)
		Khanael <i>et al.</i> (2021), O'Faircheallaigh (2015), Sankaran <i>et al.</i> (2020)
	Human health and social well-being	Hayes <i>et al.</i> (2019), Hjort <i>et al.</i> (2022), Lloret <i>et al.</i> (2022)
		Social equity and justice
Sustainability by megaprojects	Climate change	Hayes <i>et al.</i> (2019), Hjort <i>et al.</i> (2022), Lloret <i>et al.</i> (2022)
		Climate change

Source(s): Created by author

Table 3.
Themes in each
research agenda

become integral to evaluation tool like Cost-Benefit Analysis (CBA) (Van Wee *et al.*, 2003) and project objectives (Ozcan-Deziz and Zhu, 2017).

Assessment of environmental sustainability. Environmental sustainability measures are intended to minimize negative impacts on the natural environment when meeting human needs (Chiu, 2004). Tools such as Environmental Impact Assessment (EIA) are used to identify, predict, assess and mitigate potential environmental impacts (Harris *et al.*, 2009). The use of assessment tools depends on the type of megaprojects and the context, and EIA is often

combined with other tools such as Strategic Environmental Assessment and Life-cycle Assessment (LCA).

Green procurement. Green procurement involves purchasing goods, services and works with reduced environmental impact (European Commission, 2016). Increasing environmental protection requirements have made green procurement necessary, especially with recent carbon reduction measures (Kadefors *et al.*, 2021). However, due to the potential cost, owners and contractors often hesitate to adopt green practices (Ahn *et al.*, 2013; Metham *et al.*, 2022). Incentives include the incorporation of environmental performance into bid evaluation indicators and contract design, increasing the chances of winning bids, and clearer regulations (Kadefors *et al.*, 2021; Metham *et al.*, 2022; Somboonpisan and Limsawasd, 2021).

Green construction. Green construction focuses on maximizing resource conservation and minimizing negative environmental impacts through scientific management and technological progress (Liu *et al.*, 2017). Economic feasibility remains a primary concern for implementing technologies (Iligan and Irga, 2021). However, contractors may engage in unsustainable construction practices due to technological disadvantages, complex stakeholder interactions, insufficient government incentives and lack of relevant competencies and knowledge (Shang *et al.*, 2021). Strong promotion and support for green practices, stricter regulatory measures and improved government regulation can facilitate green construction (Iligan and Irga, 2021; Shang *et al.*, 2021).

4.2.2 Sustainability for megaprojects. Since the United Nations Conference on Environment and Development in 1992, awareness of sustainable development has increased significantly (Sierra *et al.*, 2016). The TBL framework, proposed by Elkington (1997), has become prevalent in the past two decades. Agenda 2 *sustainability for megaprojects* aims to deliver sustainable megaprojects that meet economic, social and environmental demands, focusing on the entire lifecycle and involving more external stakeholders. While environmental impacts and green practices remain important, new themes are emerging, such as sustainability assessment, sustainable design and sustainable procurement. Social aspects that were previously neglected are also gaining attention, including social sustainability assessment, stakeholder management, social responsibility and anti-corruption.

Sustainability life-cycle assessment. Sustainability life-cycle assessment is typically used for *ex ante* decision-making (Gibson, 2006), analytical monitoring (Glasson *et al.*, 2022) and ex-post evaluation (Yu *et al.*, 2018). Most current sustainability indicator systems are based on TBL. Scholars have proposed various sustainability-oriented appraisal methodologies, such as multi-criteria Decision Analysis (Bueno *et al.*, 2015), Integrated Life-cycle Sustainability Assessment (Keller *et al.*, 2015) and Deep Uncertainty Decision Making for complex and uncertain scenarios (Bojorquez-Tapia *et al.*, 2022).

Sustainable design. Sustainable design is crucial to achieving sustainable megaprojects (Tsai and Chang, 2012). Sustainability goals for construction, operation and maintenance and demolition are usually decided during the design phase (Xue *et al.*, 2022). Early application of visualization techniques, such as Building Information Modeling (BIM), and integration with sustainability rating systems can enhance the delivery of sustainable outcomes (Lauermann, 2020; Liu *et al.*, 2018). Various platforms and methodologies enable the assessment of alternatives based on sustainability performance indexes (Triantafyllidis *et al.*, 2018; Xue *et al.*, 2022).

Sustainable procurement. Sustainable procurement involves socially and ethically responsible sourcing, minimizing environmental impacts throughout the supply chain while adopting economically sound solutions (Chartered Institute of Procurement and Supply, 2011). However, this approach is often challenged by costs, risks and budgetary pressures (Brooks and Rich, 2016). Enabling practices include selecting large-scale companies

experienced in sustainable procurement (Brooks and Rich, 2016), fostering competitive dialogue between contractors and owners (Uttam and Roos, 2015), shaping a supportive regulatory environment, selecting appropriate procurement methods, developing the construction industry and adopting innovative approaches (Ogunsanya *et al.*, 2021).

Social sustainability assessment. Megaprojects are often controversial due to social impacts, such local communities displacement, social disorganisation and health consequences (Chau, 1995; Lerer and Scudder, 1999). For instance, Perez-Sindin (2021) reported on energy megaprojects causing social disorders and tensions between immigrants and native populations. Scholars have increasingly incorporated social considerations into sustainability assessment systems for megaprojects (e.g. Karji *et al.*, 2019; Sierra *et al.*, 2017). Common tools include Social Impact Assessment, Health Impact Assessment and Social Life-cycle Assessment.

Stakeholder management. External stakeholder management has gained attention, reflecting more of a “management for stakeholder” approach (Silvius, 2017) focusing on stakeholder perception, public engagement and acceptance. Perceptions of stakeholders significantly influence their attitudes and support, potentially determining project success (Friedl and Reichl, 2016; Li *et al.*, 2016; Prayag *et al.*, 2013). Management for stakeholder is essential for resolving social conflicts and enhancing public acceptance (Devine-Wright, 2007). Factors such as decision-making democracy, social harmony and environmental impacts contribute significantly to public participation and support (Xie *et al.*, 2017). Consistently addressing stakeholders’ concerns and needs throughout the project lifecycle is crucial for ensuring equitable satisfaction (Di Maddaloni and Davis, 2017).

Social responsibility. Megaprojects social responsibility (MSR) involves stakeholder policies and practices aimed at safeguarding the well-being of the broader society throughout the project lifecycle (Zeng *et al.*, 2015). Effective MSR is crucial for megaprojects sustainability, while poor governance in MSR can be costly (He *et al.*, 2019; Ma *et al.*, 2021; Zeng *et al.*, 2022). Establishing an integrated mechanism involving businesses, government and the public is fundamental for fostering sustainable value (Ma *et al.*, 2017). Investing in environmentally responsible practices is also a way towards good MSR performance, which may foster environmental commitment among project participants, promoting spontaneous responsible behaviour that is conducive to environmental stewardship (Wang *et al.*, 2017).

Anti-corruption. Corruption, indicative of inadequate MSR, is prevalent in megaprojects due to their size, uniqueness, government involvement and technical and organisational complexity (Locatelli *et al.*, 2017). Corruption can significantly damage projects and society (Khadim *et al.*, 2021; Locatelli *et al.*, 2017; Owusu *et al.*, 2020). Despite anti-corruption measures, the most critical barriers remain unwillingness to combat corruption, as well as political and structural barriers and psychological factors such as fear of detection (Owusu and Chan, 2019). Improved transparency of project activities and well-defined incentives can be two powerful anti-corruption mechanisms (Lehtinen *et al.*, 2022).

4.2.3 Sustainability by megaprojects. In the past decade, global awareness of sustainability crises has grown significantly. Since the introduction of SDGs in 2015, scholars have increasingly connected megaprojects sustainability with the SDGs. Agenda 3 *sustainability by megaprojects* contributes to broader societal sustainability by delivering sustainable megaprojects, expanding research focus to include the “enlarged scope” of megaprojects and a wider range of societal stakeholders, particularly regarding sustainable economic growth (SDG 8), human health and social well-being (SDG 3), social equity and justice (widely emphasized) and climate change (SDG 13).

Sustainable economic growth. Sustainable economic growth involves advancing economic development while considering social and environmental factors, fostering a positive economic cycle (G20, 2019). Megaprojects often act as catalysts for economic growth (Farhadi, 2015), such as transportation infrastructure promoting productivity, employment

and inter-regional economic spillovers (Arbues *et al.*, 2015; Chakrabarti, 2018). However, megaprojects do not always contribute to sustainable economic growth in the long run. They can exacerbate spatial-economic inequality (Zhao *et al.*, 2022), and most Olympic Games have a short-lived economic impact on host countries (Tien *et al.*, 2011). China's past pattern of infrastructure investment is also considered unsustainable, as the long-term costs may outweigh the benefits (Ansar *et al.*, 2016). Transitioning megaproject investments towards high-quality development, such as by optimizing megaproject portfolio allocation, is crucial for creating sustainable economic value (G20, 2019; Hao and Cho, 2022; Zolfaghari *et al.*, 2020).

Human health and social well-being. Human health and well-being are closely linked to ecosystems, and megaprojects need to adapt to changing health, environmental and social conditions (Annear *et al.*, 2021). By focusing on the human-environment nexus, megaproject sustainability studies aim to create integrated value in megaprojects. For example, Browne and Lowe (2021) proposed livability as a criterion to assess urban megaprojects, considering social cohesion, environmental sustainability, health and social well-being and social equity, which promotes a more integrated consideration of sustainability factors. Social value also plays a role in well-being, encompassing all necessary costs and benefits that affect the welfare and well-being of the population (Freelove and Gramatki, 2022).

Social equity and justice. Social equity and justice are emphasized in all SDGs. Social equity emphasizes equal access to public services for all, while social justice prioritizes disadvantaged groups (Tang and Gu, 2015). Social sustainability requires equitable treatment and meaningful participation for all community members (Khanal *et al.*, 2021). Tailored policy interventions addressing local needs and resource limitations can promote social equity, such as coordinated urban-rural development in China (Shen *et al.*, 2012). Distributional inequity also involves future generations, as current benefits may result in environmental and social costs for future generations (O'Faircheallaigh, 2015).

Climate change. Megaprojects can significantly contribute to climate change through greenhouse gas (GHG) emissions and can pose serious environmental risks (Lloret *et al.*, 2022; Wei *et al.*, 2021). Other megaprojects, such as the construction of large wind farms or nuclear power plants, can reduce GHG emissions slowing climate change. In general, all megaprojects are vulnerable to climate change impacts, such as operational disruptions, shorter service lives and increased reconstruction and maintenance costs (Hayes *et al.*, 2019). Assessing megaprojects' risks and vulnerabilities is vital but complex due to uncertainty, the dynamic nature and cost unpredictability (Garg *et al.*, 2015; Palin *et al.*, 2021). Building resilience and adaptability into megaprojects is vital (Palin *et al.*, 2021) but necessitates a shift towards viewing them as components of complex socio-ecological-technical systems and a paradigm shift (Bollinger *et al.*, 2014; Hayes *et al.*, 2019; Lazurko and Pinter, 2022).

In summary, research themes in the megaprojects sustainability field continue to diversify. With researchers' focus in terms of sustainability goals, megaprojects scope and stakeholders broadening, research in this area has evolved from Agenda 1 to Agenda 3, but this is not a replacement. For example, while Agenda 3 focuses on achieving a long-term, broader sustainable society, attention to the immediate and local impacts of megaprojects, as seen in Agendas 1 and 2, remains crucial. Furthermore, the evolution of megaprojects sustainability follows the that of sustainable development concepts, but it is not a simple linear matter. For instance, the focus on social issues may coincide with or even precede environmental concerns, such as the early consideration of residents' resettlement issues in large dam projects (e.g. Chau, 1995; Lerer and Scudder, 1999). The development thus reflects an intensification of the level of attention and marks a transition from micro, internal and local perspectives to macro, external and holistic views, promoting more systematic and comprehensive thinking.

5. Gaps and future research agenda of megaprojects sustainability

Megaprojects sustainability research has evolved from *sustainability of megaprojects*, to *sustainability for megaprojects*, and now to *sustainability by megaprojects*. Despite the abundance of existing studies, there remains a vast space in this area. Current theories and practices are insufficient to tackle the unprecedented sustainability challenges. Calls for sustainability transition have become widespread (Bojorquez-Tapia *et al.*, 2022; Shi and Moser, 2021; Winch, 2022; Wu *et al.*, 2022a, b) and megaprojects are a key element of the response. The urgent need for rapid comprehensive action requires fundamental changes in how megaprojects are invested in and developed (Love and Luo, 2018; Shi and Moser, 2021). Both scholars and practitioners in megaprojects management need to think outside the box (Lazurko and Pinter, 2022).

To this end, the exploration of future directions in this study will shift from a “tight” basis of existing research to a “broader” adoption of practical challenges and megatrends, specifically (1) identifying relevant research gaps in the literature; (2) generating innovative insights and implications for future directions based on the major challenges of megaprojects sustainability practices, international strategies and calls from academics. Our proposed future research agenda is not limited to a megaproject perspective, but places particular emphasis on the congruence between the sustainability of megaprojects and the sustainability of wider society.

Taking *sustainability by megaprojects* as a starting point, in this section we discuss the gaps and future directions from the perspectives of the sustainability, scope and stakeholder dimensions, to develop a future research agenda on megaprojects sustainability (Figure 5).

5.1 SDGs-orientation

The SDGs represent the UN's most extensive endeavour to promote sustainable development (Biermann *et al.*, 2022). However, in megaprojects management, the SDGs often fall short in practical application. The integration of the SDGs into megaprojects, organisations and society is insufficient, and the intrinsic relationships among the SDGs are often overlooked. These will be key focus points for future researchers and practitioners.

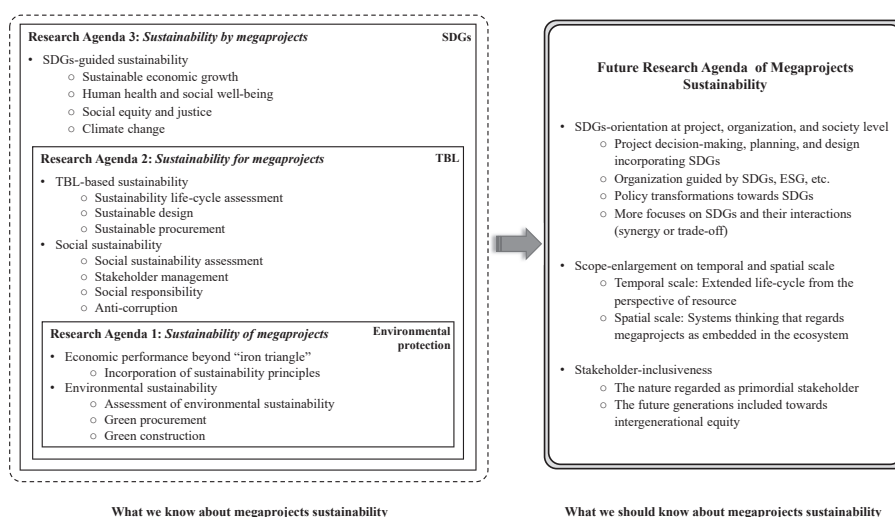


Figure 5.
The future research
agenda of
megaprojects
sustainability

5.1.1 Megaprojects level. Decision-making and design are pivotal to the sustainable performance of megaprojects. Poorly planned megaprojects can threaten the SDGs (Thacker *et al.*, 2019). Existing decision-making tools are either flawed or inefficiently applied in practice. For example, EIA has been controversial due to insufficient public participation (Pereira, 2021) and insufficient health considerations (Harris *et al.*, 2009, 2021). This finding is also consistent with the study by Thounaojam and Laishram (2022). Incomplete assessments can pose significant risks to both megaprojects and ecosystems (Lloret *et al.*, 2022). There is a pressing need for comprehensive tools to measure multidimensional sustainability in megaproject decision-making. Future research should focus on developing such tools that align with the SDGs, incorporating and enhancing existing methodologies.

5.1.2 Organisation level. Organisations actively practicing in sustainable development play a vital role in megaprojects sustainability. Together with SDGs, environmental, social and governance (ESG) matters have gained significant popularity at the organisation level (Kim and Yoon, 2023). ESG criteria are crucial to the World Bank's assessment of project risks, investment and financing decisions and project approvals. The influence of ESG extends to altering investment and development strategies in megaprojects. Despite this importance, discussions of ESG's impacts on megaproject management and governance are scant. Exploring ESG disclosure in the context of social responsibility and sustainable governance of megaprojects holds significant value (Li and Adriaens, 2024).

5.1.3 Society level. Megaprojects are often propelled by political initiatives (Flyvbjerg, 2017; Wang *et al.*, 2020). National policies are fundamental to the sustainable performance of megaprojects (Wu *et al.*, 2022a, b), especially net-zero industrial megaprojects (Geels *et al.*, 2023; Sovacool and Geels, 2021). Nevertheless, literature suggests insufficient integration of the SDGs into policy frameworks. Current efforts by government agencies and other public and private organisations towards SDGs compliance are inadequate (Biermann *et al.*, 2022; Sankaran *et al.*, 2020). Institutional fragmentation impedes the adoption of an integrated approach to sustainability (Busscher *et al.*, 2015). Critics have argued that the status quo and business-as-usual practices lead to inequitable and unsustainable development (Shi and Moser, 2021). Policy-makers must embrace a transformative approach aligned with the SDGs. Future research could explore the impact of policy changes on megaprojects sustainability or their alignment with broader sustainable development at the societal level.

5.1.4 More focuses on SDGs and their interactions. Research on megaprojects sustainability concerning SDGs remains limited (Mansell *et al.*, 2020). Existing studies focusing on individual SDGs, such as 3, 7, 8, 9 and 11, are fragmented and sparse. The nascent state of SDG integration presents numerous opportunities for future research. Important areas of focus include the interactions between SDGs and their varying impacts in regions with different sustainability levels. Firstly, actions targeting one goal may reinforce or counteract actions targeting another (Wu *et al.*, 2022a, b). Understanding such synergies and trade-offs is crucial to achieving the SDGs (Biggeri *et al.*, 2019; Fuso Nerini *et al.*, 2018) and ensuring that megaprojects' goals align with sustainable development. Second, the relationship between SDGs decouples and recouples as the country or region's sustainable development level changes (Wu *et al.*, 2022a, b). Therefore, megaprojects sustainability studies should encompass a broader range of SDGs and consider their non-linear coupling relationships with contextual factors.

5.2 Scope-enlargement

Human systems and ecosystems are intricately linked, especially in the pursuit of sustainable development (Folke *et al.*, 2016; Liu *et al.*, 2007). Megaprojects, as complex adaptive systems (Daniel and Daniel, 2019), interact with ecosystems over time and space, creating value that

extends well beyond the project lifespan. In the context of sustainability transitions, merely adjusting the status quo is inadequate; a fundamental shift away from conventional “project thinking” is imperative.

5.2.1 Temporal scale. Megaprojects delivery involves integrating substantial resources and creating value from the conceptual stage (Flyvbjerg, 2017; Flyvbjerg *et al.*, 2018; Van Marrewijk *et al.*, 2008). If megaprojects are viewed as pivot points in the resource supply and value chain, there is potential to enhance sustainability through resource recycling. Circular Economy (CE) is a burgeoning concept aimed at maximizing resource value and can prolong the megaproject lifecycle (Mignacca and Locatelli, 2021). While CE principles are not new to this domain, the research is still in its infancy (Çimen, 2021). Integrating CE with modularization in megaprojects is an innovative sustainable practice (Mignacca *et al.*, 2020). Future research should focus on implementing CE principles in national sustainability strategies, exploring the interplay between modularization and CE, and examining future perspectives, challenges and barriers. Additionally, specific plans for implementing CE in various industries and assessing the economic viability of modularization and CE in megaprojects should be pursued at national and local levels.

5.2.2 Spatial scale. Sustainability transition necessitates policy transformations in the way megaprojects work (Love and Luo, 2018). Studying the intertwined social and ecological processes of development should be “*not only within artificially established analytical borders but in the real globalized world of the Anthropocene*” (Folke *et al.*, 2016, p. 6). Future research should move beyond conventional “project thinking” and adopt a “systems thinking” approach, recognising megaprojects as components of socio-technical-ecological systems. Although systems thinking has been widely employed to analyse megaprojects complexity (Jia *et al.*, 2022; Zhu and Mostafavi, 2018), it still holds significant potential for embedding megaprojects in ecosystems for sustainability transition. It serves as an emerging tool for studying interactions among the SDGs and can improve megaprojects sustainability management by understanding the coupling between humans and ecology (Love and Luo, 2018; Swain and Ranganathan, 2021). Future research may focus on how to deploy dynamic megaprojects portfolios from the view of national system, or on megaprojects decision-making, investment and development from the view of ecosystem.

5.3 Stakeholder-inclusiveness

Inclusive development, inherent to SDGs, prioritizes equitable development within and across generations. Meanwhile, human society and nature are intertwined, co-evolving, shaping and being shaped by each other (Folke *et al.*, 2016). However, future generations and the nature are often excluded from the stakeholder groups of megaprojects. It is imperative to inclusively integrate them into future megaproject considerations but also challenging for research and practice, involving a fundamental shift in philosophy and thinking.

5.3.1 The nature. Academics have long recognised the need to consider the natural environment as a project stakeholder (Buchholz, 2004; Driscoll and Starik, 2004; Haigh and Griffiths, 2009). TBL alone is not enough; nature must be prioritized (Whyte and Mottee, 2022). The way forward should be to consider nature as a stakeholder, rather than a single-minded emphasis on considering the impacts of a project more broadly. Green infrastructure, for instance, represents an advanced practice that better integrates natural and non-natural systems, contributing significantly to infrastructure system resilience, public participation and economic efficiency (Browder *et al.*, 2019). It is more flexible and resilient than technical and social infrastructure (Mertens *et al.*, 2022). Prioritizing nature as a key stakeholder requires strong support from theoretical research and policy. Crucial aspects such as the integration of green and gray infrastructure, the costs and wider benefits, stakeholder engagement and financing all need exploration and contextualization within specific national and regional contexts.

5.3.2 The future generations. The SDGs highlight intra- and intergenerational equity. Megaprojects, being future-oriented (Whyte and Mottee, 2022), serve as legacies that will affect future generations (Flyvbjerg, 2017) for decades or centuries. However, existing studies have largely neglected to consider this stakeholder group. What are the criteria for defining future generations? How can the benefits or losses of megaprojects to future generations be defined and measured? How many generations are involved? How are the benefits and losses to current and future generations weighed? In practice, this is even more complex, as future sustainable benefits often conflict with short-term political and organisational interests (Trevino-Lozano, 2021). Addressing these questions should be a primary focus of future research. In addition, the new stakeholder theory, an evolving and emerging theory that emphasizes the value creation of stakeholders in organisations (McGahan, 2021), offers an opportunity to develop inclusive frameworks for considering future generations.

6. Conclusions

This study provides a comprehensive overview of the development, state of the art and future directions of megaprojects sustainability research, hopefully inspiring more studies and disruptive actions towards sustainability transition.

Drawing on sustainability science and project management theories, we developed a systematic analytical framework of megaprojects sustainability with three dimensions of sustainability, scope and stakeholders. On this basis, we revealed that megaprojects sustainability research has evolved from *sustainability of megaprojects*, to *sustainability for megaprojects* and then to *sustainability by megaprojects*. Research on megaprojects sustainability started simultaneously with global sustainable development. Although its overall development has lagged behind sustainable development, this field has generally evolved with the evolution of sustainable development concepts. In terms of research themes, environmental and social sustainability are most concerned, particularly topics such as climate change, environmental impact assessment, stakeholder management, public participation and social responsibility. *Sustainability by megaprojects* is the focus of future research, involving SDGs, the extended life-cycle of megaprojects and their deliverables and stakeholders on a global scale. Therefore, the future vision is to promote the sustainable development of society through sustainable megaprojects, and only megaprojects that are beneficial to the sustainable development of society can be considered sustainable.

To respond to global challenges, there remain significant gaps. Future megaprojects sustainability research is not just about refining deficiencies in current research; it requires scholars and practitioners to think outside the box, make a paradigm shift and take disruptive actions. On the basis of the framework and the state of the art, we developed a future agenda that suggests three avenues: (1) SDGs-orientation at the megaproject, organisation and society levels, with more focuses on interactions between SDGs; (2) scope enlargement at temporal and spatial scales, requiring extended life-cycles from the resource perspective and systems thinking and (3) inclusive development for stakeholder, especially regarding nature and future generations as megaprojects stakeholders.

This study contributes to the literature by revealing the overall development trajectory of megaproject sustainability research and by providing insights into future roadmaps, especially under great pressure of SDGs. Our study also provides implications for practitioners, showing them the current state of sustainable practices in megaprojects and potential focus areas for future transformation.

The study has limitations. We deliberately excluded grey literature, which may have valuable practical insights. Despite a broad and extensive selection of terms and strict adherence to SLR protocols, the screening and analysis of the literature were still subject to

the authors' subjective interpretation. Nevertheless, our focus on interdisciplinary frontiers and international reports helps to compensate for this limitation.

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